

Oswald Gyabaah

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PROFESSIONAL SUMMARY

Infrastructure engineer with 5+ years building and operating production observability platforms, Kubernetes clusters, and high-throughput data pipelines at enterprise scale. Built unified monitoring and alerting systems (Prometheus, Grafana, Elastic Stack) serving 3M+ daily requests across 800+ pods with 99.97% uptime. Experienced in distributed systems debugging, on-call incident response, and telemetry pipeline design. Automates infrastructure with Terraform, Ansible, and Python. Currently extending operational capabilities with AI-driven automation agents for SRE and QA workflows.

CORE COMPETENCIES

Observability & Telemetry Platforms

Distributed Systems Monitoring

Kubernetes at Scale (800+ pods)

Data Pipeline Engineering

Infrastructure as Code (Terraform)

Incident Response & On-Call

Metrics, Logging & Tracing

Python & Automation

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD May 2025 - Present

Manager, Finacle Services Team

- Manage the **observability stack** with dashboards tracking CPU/memory trends, transaction volumes, response times, and categorized exceptions -- enabling **threshold-based alerting** and proactive scaling across the cluster
- Manage the Finacle Services team running the on-prem Core Banking System on **Tanzu Kubernetes Grid (TKG)**, overseeing **800+ production pods** and full cluster lifecycle: deployments, patching, and **incident response**, sustaining **1.5M+ daily banking transactions**
- Lead **L2 support operations**, correlating transactional and application logs across interfaces and core banking **microservices** via the **Elastic Stack**, driving **root-cause analysis** and cross-functional remediation
- Led end-to-end QA for core banking migration: 10,000+ test cases across 13 modules and 33 applications, overseeing integration, UAT, and peripheral device validation
- Manage core infrastructure operations: debugging deployment errors, database connectivity, node optimizations, networking/storage bottlenecks, coordinating upgrades and security advisories across TKG environments

Guaranty Trust Bank Ghana LTD Apr 2024 - Present

DevOps Engineering Lead

- Deployed unified **observability platform (Prometheus, Grafana, Elastic Stack, Sentry)** for centralized **logging, tracing, and real-time alerting** -- accelerating root-cause analysis and improving mean time to resolution across **distributed systems**
- Architected high-availability **Kubernetes clusters** across multiple on-prem environments handling **3M+ daily requests** with **99.97% uptime SLA**; implemented auto-scaling and self-healing mechanisms
- Implemented CI/CD automation cutting deployment time by ~90% and increasing release velocity ~5x; introduced GitOps with FluxCD and integrated Veracode and Trivy scanning for automated security analysis
- Automated 95% of infrastructure configuration management using Ansible, **Python**, and Bash scripts -- transforming multi-day manual processes into reproducible, version-controlled deployments
- Implemented stateless, version-controlled cluster deployments with FluxCD, Helm, and Azure DevOps for consistency, idempotency, and rapid disaster recovery
- Founded the bank's DevOps team, leading engineers to establish modern delivery practices -- introducing **Kubernetes, containerized** deployments, disciplined version control, and automated release workflows

Chief Technical Officer

- Developed a **high-throughput data processing pipeline** handling **1M+ daily jobs** (payments, messaging, notifications) with SMS, email, and payment provider integrations including failover and provider fallback
- Architected full platform on AWS orchestrating **containerized microservices** using EKS and ECS with autoscaling, load balancing, and seamless inter-service communication
- Designed and implemented disaster recovery: parallel GCP production environment mirroring AWS using GKE, Cloud SQL, and Cloud Storage with secure VPCs, IAM policies, and dual-environment CI/CD
- Automated **infrastructure** provisioning with **Terraform**, reducing setup time by 90% while building CI/CD pipelines to streamline deployments
- Reduced peak latency by 90% through database optimizations: query tuning, caching layers, and indexing

PROJECTS**Virk Cloud -- virk.cloud** Founder & CTO

Production-grade **Kubernetes** PaaS enabling startups to deploy clusters instantly -- fixed billing, built-in **observability** stack for centralized **monitoring**, real-time log streaming, and **alerts**. Full **infrastructure** on VPS: cluster deployment (RKE2), automated with FluxCD and **Terraform**.

RKE2, FluxCD, Terraform, Prometheus, Grafana

AI Agents for Infrastructure Operations Current

Building specialized AI agents for QA automation and sysadmin operations on production Linux servers. Developing agent-driven SaaS products leveraging RAG, tool-use, and multi-agent orchestration patterns for **infrastructure** automation.

Python, LLM APIs, RAG, Multi-Agent Systems

Education Research and Innovation Foundry -- erifoundry.org Founder & Research Lead

Leading design of AI-driven research initiatives in adaptive learning, cognitive modeling, and educational analytics. Prototyping applications of LLMs in human-centered learning systems.

Python, LLMs, ML Research

EDUCATION**BSc Computer Engineering -- Ashesi University**

Sep 2016 - May 2020

Berekuso, Ghana

CERTIFICATIONS

Certified Oracle DevOps Professional -- **Oracle**

Certified Oracle Cloud Solutions Architect Associate -- **Oracle**

AI Engineering Specialization (Agents, RAG, LLM Applications) -- **ByteByteAI**

SKILLS

Observability & Telemetry: Prometheus, Grafana, Elastic Stack (Elasticsearch, Logstash, Kibana), Sentry, Metrics/Logging/Tracing Pipelines

Cloud & Infrastructure: AWS, Azure, GCP, OCI, VMware, On-Premise/Bare Metal Data Centers, Linux Administration

Container Orchestration: Kubernetes (EKS, GKE, TKG, RKE2), Docker, Helm, Istio Service Mesh

IaC & Automation: Terraform, Ansible, Python, Bash, GitHub Actions, ArgoCD, FluxCD, Azure DevOps

Networking & Security: IPSec/OpenVPN, TLS/SSL, WAF, OAuth2, Data Sovereignty Compliance

AI/ML & Agents: LLM Integration, RAG Systems, AI Agent Development, Multi-Agent Orchestration